Sahaj Bhadja

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Summary

Systems-focused engineer experienced in **C++, Python, Linux, concurrency, performance analysis, distributed execution, and ML workloads**, building reliable software across simulation, data-processing, and production environments.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Operating Systems, Analysis of Algorithms, Database Systems, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Computer Networks, Data Structures, Object-Oriented Programming, Software Engineering

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Reworked asynchronous Python orchestration with **explicit job states, concurrent workers, robust failure isolation, and production diagnostics**, tripling throughput for 1K+ monthly jobs while sustaining 99%+ application reliability.
- Optimized multi-tenant production data access through **indexed queries, normalized schemas, and resource-aware profiling**, reliably reducing dashboard latency from 1.2 seconds to 350 milliseconds across 500+ daily users.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Developed multithreaded Python ingestion for PDF, DOCX, ZIP, OCR, and multimodal workloads, reducing turnaround 60% through **bounded concurrency, fault-tolerant execution, secure workload validation, and production performance testing**.
- Diagnosed and resolved production failures across Flask services, model pipelines, and cloud data stores, supporting 10K+ daily requests with **structured logging, automated tests, CI/CD, and reproducible debugging**.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, CA*

- Built Python benchmarks for 100K+ simulated driving trajectories, improving planning 18% while applying **batch execution, numerical analysis, instrumentation, controlled experiments, failure analysis, and repeatable ML workload evaluation**.
- Implemented a spatial-temporal encoder combining HD-map geometry, lane context, and ego trajectories with multi-head attention, improving closed-loop stability 15% through **profiling, controlled benchmarks, and systematic failure analysis**.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) *India*

- Designed object-oriented services and distributed data pipelines across Python, AWS Glue, S3, and SQL, improving response times 30% through **modular interfaces, resource profiling, asynchronous processing, and bottleneck removal**.
- Automated 20+ unit and integration suites across Windows and Linux virtual machines using PyTest, Mockito, and Jenkins, increasing pipeline efficiency 68% through **repeatable builds, regression isolation, and diagnostics**.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) *India*

- Built eight modular services and real-time WebSocket flows that synchronized 2M+ records, applying **event-driven communication, shared schemas, failure monitoring, and reliable state propagation** across distributed application components.
- Automated Docker and Jenkins deployments, reducing release time 50% while using **Linux containers, Sentry diagnostics, structured logs, integration testing, and rollback-ready builds** to improve production operability and troubleshooting.

Skills

Systems: C++, Python, C, Java, Bash, SQL, Linux, POSIX, Processes, Threads, Concurrency, IPC, System Calls, Kernel Modules, Drivers
Networking: TCP/IP, UDP, RDMA, RoCE, InfiniBand, MPI, NCCL, UCX, Sockets, Distributed Systems, Fault Tolerance
Performance & ML: GDB, perf, Valgrind, Profiling, Benchmarking, Latency, Throughput, PyTorch, NumPy, Multiprocessing, Data Structures, Algorithms
Engineering: Git, Docker, Kubernetes, Jenkins, CI/CD, Unit Testing, Integration Testing, Debugging, Observability, Code Reviews

Projects

Linux Accelerator Runtime Lab | *C++, Python, Linux, POSIX, ioctl, mmap, epoll, GDB, perf*

- Built a **Linux accelerator runtime and character-device interface** with memory-mapped command queues, asynchronous completion handling, resource lifecycle management, fault injection, Python bindings, GDB workflows, and perf-based latency profiling.

Distributed Collective Transport Profiler | *C++, Python, MPI, RDMA, RoCE, Linux*

- Developed a **multi-node communication profiler** for collective workloads, measuring latency, bandwidth, queue depth, and synchronization stalls while comparing MPI transports and diagnosing cross-node bottlenecks through reproducible benchmark traces.

CPU Scheduling Simulator | *C++, React, JavaScript* | Live

- Implemented performance-sensitive C++ algorithms for **FCFS, SJF, SRT, Round Robin, priority, and multilevel queues**, validating correctness while exposing execution timelines, waiting-time tradeoffs, and starvation behavior through interactive controls.